

Subtract 00:04:00 from timestamps to find place in audio/video recording.

1

00:04:20.760 --> 00:04:21.660

Krzysztof Paruch: Huh.

2

00:04:24.140 --> 00:04:25.009

Jamsheed Shorish: Chris, how are you?

3

00:04:25.590 --> 00:04:28.240

Krzysztof Paruch: Hey, I'm fine. Hey, how are you, Jumpshit?

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00:04:28.830 --> 00:04:29.910

Jamsheed Shorish: Doing pretty well.

5

00:04:30.800 --> 00:04:31.500

Krzysztof Paruch: Yeah?

6

00:04:32.150 --> 00:04:37.020

Jamsheed Shorish: Well, good... glad to hear that you're still with us for future Oh, yes.

7

00:04:37.220 --> 00:04:41.989

Krzysztof Paruch: Yes, I mean, like, if you don't kick me out, I mean, like, I'm gonna come back.

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00:04:41.990 --> 00:04:43.900

Jamsheed Shorish: No, I don't think... I don't think we're gonna kick you out, no.

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00:04:44.400 --> 00:04:53.889

Jamsheed Shorish: I think it's been great. I think it's good to be able to continue moving on the same track, because it's good to have a nice, unified touchpoint that you're able to build upon each time you give another talk, so... super cool.

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00:04:53.890 --> 00:05:07.230

Krzysztof Paruch: Yeah, yeah, and it's very interesting, like, to have this group. I do have other groups as well, where I talk about, kind of, like, more implementation-related topics, but, like, this group is

amazing at, giving feedback on the economics and,

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00:05:08.050 --> 00:05:10.550

Krzysztof Paruch: And operation research side of things.

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00:05:10.550 --> 00:05:11.120

Jamsheed Shorish: Cool.

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00:05:11.580 --> 00:05:24.280

Jamsheed Shorish: Yeah, nice. And thanks also for passing on the papers. Those are actually quite interesting. In fact, even the last one you put together, it seems like they're really pushing the envelope here on trying to grab as much as they can on evaluating and making a choice between these different frameworks.

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00:05:24.890 --> 00:05:30.389

Krzysztof Paruch: Yeah, did you have a chance to look into them? I mean, I was wondering, I was basically betting on you, like, to at least have a peek.

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00:05:31.050 --> 00:05:33.110

Jamsheed Shorish: Yeah, I had a peek. Yeah.

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00:05:33.130 --> 00:05:42.780

Jamsheed Shorish: Yeah, at a peak, I made, again, like, I had this little, like, small summation at the end that maybe we can go through, literally just, like, 5 minutes of tying some stuff together that,

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00:05:42.780 --> 00:05:57.979

Jamsheed Shorish: helps, in essence, kind of place each one of these within very broadly defined categories, but I'm certainly interested to hearing when you speak today upon which one you felt was the most persuasive, because they each use a completely different idea on how they want to make the assessment on evaluating prediction markets, so...

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00:05:58.750 --> 00:06:11.930

Krzysztof Paruch: Right. I mean, I'm coming from a different angle, so, like, I'm trying to just... because I started doing this evaluation that I showed you last time, where I just, like, you know, like, pushed signals, like, live market signals into the market, and...

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00:06:11.950 --> 00:06:20.869

Krzysztof Paruch: wanted to compare, how it... how both of them perform, and then I just kind of, like, took a step and said, hey, hey, actually, how do you do this properly, and, like, what exists out there, and...

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00:06:20.870 --> 00:06:35.489

Krzysztof Paruch: Interestingly, because, like, the Karimi paper is pretty recent, so, like, there seems not to be anything similar to what I'm doing, like, which is good news, like, this is, like, closing the research gap, but at the same time, yeah, there are...

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00:06:35.580 --> 00:06:40.679

Krzysztof Paruch: Already, like, there's a handful of approaches that you can leverage or discuss.

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00:06:41.000 --> 00:06:54.450

Krzysztof Paruch: to take this comparison, right? But, I'm still kind of, like, trying to deduct an implementation recommendation, but what it seems to offer, kind of like the analysis that I've done so far, is... is more,

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00:06:54.900 --> 00:07:12.520

Krzysztof Paruch: That there is a sweet spot in between where you would basically have some markets perform better on lower trading activity, or, like, on thin markets, and you have a better performance of other market types, the order bookmarket type, the central... the double auction.

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00:07:12.570 --> 00:07:21.769

Krzysztof Paruch: for higher... for higher activity. And this seems to be... this seems to be the takeaway. So, like, either, like, recommending some hybrid solutions.

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00:07:21.870 --> 00:07:25.180

Krzysztof Paruch: Like, how you can transition between those markets.

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00:07:25.800 --> 00:07:41.660

Krzysztof Paruch: And I know that, kind of, like, batch auctions are probably, kind of, like, the thing that you would do here, right? Which is some of the AMNs, like CowSwap, for example, they are... they're doing exactly this. So, like, they are collecting trades, and then they batch off, kind of, like, the...

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00:07:41.700 --> 00:07:47.570

Krzysztof Paruch: The, the, the, the matching, between different, solvers.

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00:07:48.100 --> 00:08:12.309

Krzysztof Paruch: So this seems to be kind of like the, yeah, at the bottleneck, this seems to be the, this seems to be the solution to go, implementation-wise. But for the edge cases, the answer is pretty obvious. So, like, if you want to employ a market with 5 participants, like an expert who, Paul, in your company, like a business case, for example, where you say just, okay, let's... let's build the bridge market for

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00:08:12.440 --> 00:08:19.520

Krzysztof Paruch: Company information aggregation, and you really want to do it, like, there seems to be a clear recommendation towards the market scoring rules.

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00:08:20.080 --> 00:08:39.989

Krzysztof Paruch: And if you want to go retail and target, kind of, like, you know, every American to vote on the president, then you would go into the order booked kind of things, where the big polymarket calci type of prediction markets tend to converge to, because they want to have, kind of, like, public visibility and, you know, a huge hype.

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00:08:41.970 --> 00:08:42.919

Jamsheed Shorish: Nice.

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00:08:43.039 --> 00:08:45.409

Jamsheed Shorish: Well, great, I think we are...

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00:08:45.410 --> 00:08:45.800

Krzysztof Paruch: Bye.

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00:08:45.800 --> 00:08:49.780

Jamsheed Shorish: About... beginning, we got a few people here, maybe we...

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00:08:50.150 --> 00:08:54.359

Jamsheed Shorish: Yeah, why don't we go ahead and get started? We'll see if anybody else drops in, as we move forward.

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00:08:55.550 --> 00:09:07.350

Krzysztof Paruch: Yeah, so you caught me in the midst of the prologue, but, I can... yeah, I can basically just, start sharing my screen and,

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00:09:08.590 --> 00:09:18.989

Krzysztof Paruch: I guess that the best approach would be to just go through the whole motivation of it, like, step by step. So for you, Jamsheet, like, the initial was just, like, the summary, because you are very deep into the weeds of what I'm doing, but...

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00:09:19.880 --> 00:09:28.000

Krzysztof Paruch: For everyone else, I think we should need to take a step back, and now... yeah, okay.

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00:09:29.090 --> 00:09:34.330

Krzysztof Paruch: There is a huge hack, MD, I don't know if you're gonna... if you're going to be able to go through it.

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00:09:34.590 --> 00:09:47.220

Krzysztof Paruch: It really depends how we want to make the use of the time. We have Jim Sheet, 5 to 10 minutes at the end to bridge this off towards some modeling approaches, especially system modeling.

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00:09:47.390 --> 00:10:12.019

Krzysztof Paruch: approaches. But my motivation to do this is, yeah, to evaluate, basically, performance of different market-type implementations, so looking closely at the micro-market structure. We've been talking about prediction markets and information aggregations for, I think, over a year now. So, so this is something that, we would be doing here, just, taking off from this research stream.

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00:10:12.500 --> 00:10:22.509

Krzysztof Paruch: With two ambitions in mind. So, first of all, kind of, like, find an answer on when... which market type performs better, like, when do you select which mechanism, and how to parameterize it.

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00:10:22.980 --> 00:10:36.060

Krzysztof Paruch: And the other, which we are not going to answer

today because there's nothing in the literature so far that I've found that is answering this, but obviously, like, what is the proper implementation architecture for realizing this?

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00:10:36.230 --> 00:10:50.230

Krzysztof Paruch: So, talking about all of the topics that we've touched upon over the last couple of months already, which is basically the full stack of implementation, prototyping this into a functioning application.

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00:10:50.370 --> 00:10:58.530

Krzysztof Paruch: But also everything related to Oracles, resolution of those contracts, setting up of those contracts, governance of those contracts, and so on and so on.

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00:10:59.650 --> 00:11:08.639

Krzysztof Paruch: So What we aim to do here is, we... we are asking what the performance comparison,

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00:11:08.900 --> 00:11:24.429

Krzysztof Paruch: is, and how to assess it, actually, in the first place, between different prediction market mechanisms, and we select two of them. So, like, there are many, many more that you could deploy, but those are the most prominent in both in literature, but also in... actually in practice.

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00:11:24.760 --> 00:11:27.899

Krzysztof Paruch: The parameterizations of those mechanisms.

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00:11:27.900 --> 00:11:50.499

Krzysztof Paruch: and implementation system architectures, right? On-chain, off-chain components. We will not talk about this. We will talk a little bit about those two things, but mostly focus on the first thing, because parametrizations are mostly explored in dedicated research streams that focus only on one of those two mechanism families. So, there is a full

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00:11:50.810 --> 00:12:12.849

Krzysztof Paruch: branch of research that says, okay, let's look into CDA markets, and then let's see, like, how to parameterize it and how it performs. Let's look into MSR markets and look into parameterizations. But what the ambition of this conversation today is to find recommendations on when to choose one over the other.

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00:12:13.710 --> 00:12:17.089

Krzysztof Paruch: So, so this is what you want to do. And,

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00:12:18.040 --> 00:12:29.040

Krzysztof Paruch: one of the main ambitions is to actually assess what a meaningful performance comparison could be. So, like, what is it, and how do you do it, and what the approach is.

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00:12:30.240 --> 00:12:48.599

Krzysztof Paruch: So let's look at what I found in the literature over the last couple of weeks, and what you... what you see happening a lot is that there are three different ways how you can compare those mechanisms. You can do it ex-ante, ex-interim, and X post.

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00:12:48.820 --> 00:12:54.640

Krzysztof Paruch: And we will, be talking about two of them today. Not all of them, but two of them.

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00:12:55.240 --> 00:13:07.660

Krzysztof Paruch: Simply because we are not interested in the realized performance, but we are interested in the accuracy of the forecasts, and we are interested in the information aggregation process.

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00:13:07.660 --> 00:13:17.880

Krzysztof Paruch: So this is why we will be talking about markets that have resolved, and markets that are still ongoing, and see which information you can draw

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00:13:17.880 --> 00:13:21.610

Krzysztof Paruch: From observing the data that you can see there.

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00:13:21.970 --> 00:13:41.399

Krzysztof Paruch: And we will do so by using different methods. So what you see is, like, in practice, the employed are experimental methods. So you have either live experiments, so laboratory experiments, or you have experiments on actual employed systems.

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00:13:42.020 --> 00:14:01.779

Krzysztof Paruch: You have pure simulation approaches, where you basically just spin up a modeling framework, an agent-based modeling

framework, for example, and then you just, you know, ingest traders with different strategies and see how this plays out. I mean, obviously, you can... what is happening, actually, in one of the publications that we will discuss.

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00:14:01.780 --> 00:14:15.010

Krzysztof Paruch: You can parametrize those agents, or you can set up, you can validate, micro-validate the model by using actual data from previous experiments, or from real-life observations.

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00:14:15.050 --> 00:14:20.699

Krzysztof Paruch: But still, the simulations that are happening there are pure numerical experiments.

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00:14:21.100 --> 00:14:25.280

Krzysztof Paruch: Or you can do econometric analysis, and we will do so today as well.

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00:14:25.680 --> 00:14:39.909

Krzysztof Paruch: So, so let's first map out those evaluation approaches, and see how they can be compared. And, when we look at the ex-ante evaluation, this is the framework that is listed here first.

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00:14:40.090 --> 00:14:47.910

Krzysztof Paruch: The focus is forecast accuracy, so the question that we're trying to answer is which mechanism produced the most accurate forecast?

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00:14:48.120 --> 00:15:02.760

Krzysztof Paruch: So you're basically observing what's happening, comparing against the result, and then see what correctness or accuracy of the forecast was, which obviously requires the event to be resolved.

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00:15:03.250 --> 00:15:27.850

Krzysztof Paruch: And it requires the final probabilities plus the ground truth. And typically, you would do this with scoring rules, so you have the Brier score or the log score, where you can compare a kind of, you know, forecast against the realized outcome. And those values go between 0 and 2, because you can... if you're super correct, like, if you say it will happen and it happens, then you get a full score.

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00:15:27.850 --> 00:15:37.720

Krzysztof Paruch: And if you, bet on the opposite, so like you say, it will not happen, but it happens, you will get zero, and everything in between is on the scale.

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00:15:37.920 --> 00:15:40.459

Krzysztof Paruch: On the spectrum between those two numbers.

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00:15:40.900 --> 00:15:44.559

Krzysztof Paruch: What you have as next is the ex-interim Evaluation.

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00:15:44.750 --> 00:15:53.880

Krzysztof Paruch: So this is looking at the information aggregation process, and we will also have a publication today that we will discuss, which is...

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00:15:53.910 --> 00:16:09.259

Krzysztof Paruch: I'm trying to answer the question, which mechanism incorporates information faster? So, what is happening here is we have two markets that run in parallel. They are distinct from each other, so that they are... that they are,

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00:16:09.750 --> 00:16:19.899

Krzysztof Paruch: Enabled by a different provider, but still, the question is the same that is trying to be answered, and then you have arbitrage between those number... those markets.

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00:16:20.060 --> 00:16:38.830

Krzysztof Paruch: arbitrage opportunities. So, there is, there is, some correlation between those, those time series. And what you can now investigate is when new information arrives, which of those markets incorporates this information faster.

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00:16:39.210 --> 00:16:46.539

Krzysztof Paruch: So, this doesn't require resolution, because you are doing this on an ongoing live market.

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00:16:46.740 --> 00:16:55.440

Krzysztof Paruch: But you need the price time series during trading. And I think what I have here in this table is I do have another column. So, yeah, exactly, so it should go...

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00:16:55.750 --> 00:17:07.290

Krzysztof Paruch: So it should go here, and we have some example literature listed here as well, and, yeah, the interpretations, which is, like, it measures information efficiency during the active trading period for this mark.

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00:17:08.359 --> 00:17:29.859

Krzysztof Paruch: Okay, so this is X under X interim, and now we move to X post evaluation, where we just observed the realized market performance, so how well did the market perform overall? I didn't look into this at all, so I cannot tell you much about this, but this is what you, what you would be needing to answer this question.

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00:17:29.860 --> 00:17:54.719

Krzysztof Paruch: And, you look at price deviation, welfare, trader, profit, stuff like that, which is not interesting to answer my question first, and B, it also... like, I didn't find any proper reference that is targeting this. There is only one paper which I believe focuses merely on... on one of the two mechanism families. So, not the comparison that we wanted to be looking for.

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00:17:56.030 --> 00:18:04.410

Krzysztof Paruch: And then what you have is simulation laboratory experiments, so the question is, how does the mechanism perform under controlled conditions?

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00:18:04.410 --> 00:18:17.769

Krzysztof Paruch: And you can do everything, basically, that you want, so you can set up some form of an modeling environment, you can fine-tune, parameterize the model, and then you can run experiments and see how they behave.

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00:18:19.100 --> 00:18:24.820

Krzysztof Paruch: Okay, any question at this stage? Maybe comments? I don't...

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00:18:30.340 --> 00:18:38.769

Krzysztof Paruch: Cool. Okay, so what do we have here? We have a few papers that we've been looking at. We... it...

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00:18:38.980 --> 00:18:47.689

Krzysztof Paruch: maybe it doesn't really bring much more additional information if we go through this table. So we have definitely this one, Atanasov et al. This is the one that I shared

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00:18:47.690 --> 00:18:59.690

Krzysztof Paruch: yesterday, in the evening. So we won't have time to discuss it, because I just simply didn't have the time to summarize it for the presentation, but maybe we can just leave it, as kind of like an open discussion at the end.

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00:19:00.050 --> 00:19:24.529

Krzysztof Paruch: But what they do is they set up a large experiment where they randomly assign people towards two markets. So, like, they let a group of people trade on a CDA market, and one group of people to trade on a MSL market, and then they just see how this market performs, and they compare the forecast accuracy.

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00:19:24.830 --> 00:19:41.110

Krzysztof Paruch: So, this is basically one of the few situations where, there was actually an experiment where both markets were active at the same time, and, people have been betting on both of them, although,

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00:19:41.180 --> 00:19:45.149

Krzysztof Paruch: There was no single trader that was participating in both of the markets, right?

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00:19:47.220 --> 00:20:02.050

Krzysztof Paruch: We have Karimi and Dimitrov, which is a 2025 paper. I've seen some preprints, 23, but I think the official publishing date is 25. What they do is this ex-interim approach, so, like, this here, they focus on the process.

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00:20:02.360 --> 00:20:15.689

Krzysztof Paruch: And that they don't need the market to be resolved, and what they do is they try to see how new information affects market prices, and which of the markets incorporates this information faster.

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00:20:15.690 --> 00:20:39.600

Krzysztof Paruch: So, what they find is that there is a certain threshold until which the scoring rule market incorporates information faster compared to CDA markets, and then this reverts, and then after this, above this threshold, the other market performs better, which is

something that is contrary to the belief or the, you know, results discussion in literature.

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00:20:39.600 --> 00:20:42.450

Krzysztof Paruch: Because, here.

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00:20:42.690 --> 00:20:51.950

Krzysztof Paruch: There was this common belief that, above this threshold, both markets are fine and do well, comparatively well.

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00:20:51.970 --> 00:21:03.209

Krzysztof Paruch: But this appears not to be the case. So they are actually able to reject the hypothesis that both markets perform equally well above the threshold, and there is a clear winner, like, above

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00:21:03.500 --> 00:21:05.620

Krzysztof Paruch: we have the CDA mock.

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00:21:07.100 --> 00:21:30.459

Krzysztof Paruch: And then we have Klinger and Meyer, 2016 paper, with also some preprints, 2013, and even before, where they do a simulation comparison, set up a multi-agent simulation, calibrated with lab data, and the interesting part about this lab data is that it's coming directly from a lab experiment from Robin Hansen of 2006.

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00:21:30.460 --> 00:21:38.120

Krzysztof Paruch: So they basically... They play into the same direction by trying to pick up where

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00:21:38.120 --> 00:21:54.489

Krzysztof Paruch: Hansen apparently picked off with kind of, like, his laboratory experiments, trying to evaluate which of the mechanisms perform better, the one that he proposes, the market scoring rule, or the one that exists for decades already, which is the continuous double auctions.

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00:21:54.490 --> 00:22:09.110

Krzysztof Paruch: And they did it in a laboratory setting. Hanson did it in a laboratory setting, but Klinger and Meyer just spun up a simulation environment and run a huge battery of tests

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00:22:09.110 --> 00:22:20.480

Krzysztof Paruch: To see how those two mechanisms perform in different circumstances and under different, trading, behavior and population compositions.

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00:22:22.000 --> 00:22:36.039

Krzysztof Paruch: Okay, then this section discusses all the other work that I've rejected for today's presentation, because they are simply not focusing on what, we want. So, like, there are some things that are good, but, like, not really fully, so what I'm trying to

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00:22:36.040 --> 00:22:45.349

Krzysztof Paruch: just showcase here is, all of the approaches, the ex-ant exposed, ex-interim, and all of the simulation experiments, the laboratory experiments, all of this exists.

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00:22:45.350 --> 00:23:08.710

Krzysztof Paruch: But I was only able to find three distinct papers that focus on the problem, or the problem formulation that I have defined, which is exactly comparison between two mechanism families and the performance evaluation. So the question now is, how do you actually define a metric? And this might be the first stopping point for a discussion, because everything else beyond

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00:23:08.710 --> 00:23:27.559

Krzysztof Paruch: would then be the discussion of those papers. But the first attempt of today's conversation is actually to find some normative definition of what it is actually, given the information that we have here, what it is actually that you can do to define some form of an evaluation or comparison framework

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00:23:27.560 --> 00:23:30.709

Krzysztof Paruch: And how you can, you can learn,

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00:23:30.710 --> 00:23:37.489

Krzysztof Paruch: You can learn from that, and yeah, use it properly for, yeah, anything that you want to do.

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00:23:38.710 --> 00:23:41.580

Krzysztof Paruch: Maybe let's stop here, maybe there are some comments.

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00:23:43.860 --> 00:24:02.850

Jamsheed Shorish: So, we talked a bit last time about the idea of, like, what is the prediction market for? Like, who is it for? And I'm pre-jumping a little bit towards some of the summation, like, commentary that I'll have at the very, very end, but just to open things up, one of the questions that you can make is a quick dividing line here between

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00:24:03.150 --> 00:24:10.689

Jamsheed Shorish: The individuals who stand to benefit from using the prediction market versus the individuals who stand to benefit from creating

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00:24:10.800 --> 00:24:28.569

Jamsheed Shorish: the prediction market. We had that distinction between the two individuals before, because there are some characteristics of auctions, you know, in particular, very simple auctions without frictions. There's this revenue equivalence where you can actually not have to worry about the design of the auction, because you'll be able to get, as a revenue maximizer, if you're creating one, the same revenue if you're a seller.

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00:24:28.570 --> 00:24:33.950

Jamsheed Shorish: These things get, of course, blocked by frictions and different types of implementations, so one thing you might

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00:24:33.950 --> 00:24:37.839

Jamsheed Shorish: Come to mind is, how do you select the proper criteria

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00:24:38.120 --> 00:24:42.970

Jamsheed Shorish: For the use case that is built around the question, Who's this for?

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00:24:43.150 --> 00:25:00.880

Jamsheed Shorish: And I think once you've answered that question, then there are natural things that will eventually come out of that, either derivative or kind of coming from the analysis that you've already presented... you're going to be presenting or have been presenting from the papers that are there, or we start to think about that from, you know, purely from an engineering point of view. If we build a system, what's it supposed to do?

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00:25:01.180 --> 00:25:07.830

Jamsheed Shorish: So, I'll start there. It's a bit provocative, but I just want to kind of get some feeling for, like, us sort of brainstorming on that on the basis of what you just put together.

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00:25:09.060 --> 00:25:09.740

Krzysztof Paruch: Sure.

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00:25:10.140 --> 00:25:15.109

Krzysztof Paruch: I would like to go last, like, if there are other people who have comments first, I don't want to reveal too much.

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00:25:16.520 --> 00:25:36.239

Zargham: I'll add the kind of... this is extending the question rather than answering it. I think also prediction markets have non-participant stakeholders, insofar as this is more closely related to the people who are launching them, but you have the layer where people are launching and expecting to benefit, you know, people who are participating, and

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00:25:36.240 --> 00:25:55.669

Zargham: at least expecting to benefit, but then you have people who are consuming it as an information source, or as a kind of pricing oracle, where the pricing oracle in term is... turn is, like, using it to estimate the probability, or the, like, the thing that is being priced. And so, you know, I think the person who's launching the market cares about

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00:25:55.670 --> 00:26:10.699

Zargham: presumably both making money themselves, and plausibly, the value proposition there relates to both of those other stakeholders, the class that participates in the market, and the class that uses the market as an oracle. If that makes sense, I'm just...

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00:26:10.700 --> 00:26:29.799

Zargham: breaking it out into three groups, because I think you do kind of need the third group in order for this to not be... like, a prediction market isn't just self-serving economic mechanism if it's providing the source of an oracle for facts, which in turn affects behavior, and that closes the loop in a wider arc.

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00:26:30.650 --> 00:26:34.009

Krzysztof Paruch: Right. I mean, like, you can call those people free riders, obviously, because they have.

122

00:26:34.010 --> 00:26:34.580

Zargham: I don't...

123

00:26:34.960 --> 00:26:59.370

Zargham: That's right, though, that you're not a free... and don't get me wrong, like, I get why you would say that, but I'm also being explicit that they're beneficiaries of the information production, which may or may not end up being material to the economy, the market that it closes, the wider one, in ways that are not free-rider. Because if they then, in turn, spend time, money, resources on activities.

124

00:26:59.370 --> 00:27:22.119

Zargham: informed by the information flow, then consuming information and producing something else is not free ridership, as long as it closes through the larger model of the... again, not the market, but the system the market is a subsystem of may have valid, economically stable feedback loops for which it would be unfair to characterize information consumers as free riders.

125

00:27:23.530 --> 00:27:34.040

Krzysztof Paruch: Okay, yeah, I mean, like, what I had in mind by saying that was just, like, that they don't have to pay, but learn something for it, right? Like, they have the beneficiaries of the information revelation, and, yeah.

126

00:27:34.040 --> 00:27:57.880

Zargham: From the perspective... well, part of what I'm trying to emphasize is the system... like, you know, we talk about systems architecture as the sandwich model, right? The system of interest, this one, is the market design. The market design has a system below it, which is what we're talking about, how it's made, and, like, who participates in it. But it sits inside of another system, and so when you do the sandwich model, you're always looking at the system of interest.

127

00:27:57.880 --> 00:28:22.230

Zargham: in terms of the system it's inside of, and the system that makes it up, and I'm just highlighting the system it's inside of, because we've been entirely discussing how it's made up side of the sandwich, and the system it's inside of has, consumers of that

information, which, again, from the perspective of the market, you could call them free riders, because they're consuming that information and not paying back into it, but from that outer system.

128

00:28:22.230 --> 00:28:47.100

Zargham: It's not necessarily an appropriate assumption to just cast them as free riders, because the overall effectiveness and stability of the system with the market, with this market in it, actually requires them to take that information and do something useful with it in a way that kind of creates value in the world, and ideally feeds back on benefits to potentially the people launching or participating in the market. So if the loop closes adequately.

129

00:28:47.100 --> 00:29:05.550

Zargham: you can get a stable system one level out, which in my mind is a prerequisite for the long-term stability of the subsystems. If you can't create a stable, larger loop system that makes sense economically, then you're not going to expect the persistence of the subsystem, in this case, the market we're discussing.

130

00:29:06.240 --> 00:29:25.850

Krzysztof Paruch: Love it. You always come up with a meta model of everything, which is cool. It fits perfectly in the discussion. I mean, one thing that comes to mind immediately when you think about it is, like, the problem that we're not going to touch upon today, but there is this whole funding problem of those markets, or of those information revelation mechanisms, where you just say, okay, there is someone needed to subsidize a market.

131

00:29:25.850 --> 00:29:32.939

Krzysztof Paruch: Actually, one of the papers is called To Subsidize or Not To Subsidize, which is how they base their evolution on, like, which mechanism is better.

132

00:29:32.940 --> 00:29:44.669

Krzysztof Paruch: But there is some overarching interest that is obviously true, that goes beyond the whole micro-market structure and the interactions of agents on top of that.

133

00:29:44.670 --> 00:29:55.440

Zargham: And the thing I'm trying to say here is not that we should go meta, meta, meta. What I'm saying is that anytime there's a particular system of interest, you can sandwich it with boundary conditions in scale.

134

00:29:55.440 --> 00:30:19.089

Zargham: So the scale one meta above it characterizes the landscape in which the thing is being designed, and it has requirements insofar as it is complementary to, useful to, or is situated within a larger system, but that's just one meta up, and then that's how you situate and evaluate whether you're successful, and that gives rise to useful evaluation criteria, and there's, like, the one meta down.

135

00:30:19.090 --> 00:30:43.549

Zargham: which is going into the design. And you can't actually get a long-term, sustainable, stable system in the actual world unless you solve what is effectively the system of equations produced by what's internally feasible to manufacture, to build, to operate, and maintain, inner picture, and then what is useful, integrated, connected, meeting real needs, and essentially closing a loop within the system outside of it.

136

00:30:43.550 --> 00:31:02.050

Zargham: which allows that effort to be funded, sustained, and otherwise, you know, deemed worthwhile in the world, interconnected with larger systems. But it's a sandwich model because you go down one click and up one click. You don't want to drill deeper than one click down, because then you're getting into the weeds, and you don't want to drill one

137

00:31:02.050 --> 00:31:21.160

Zargham: deeper... further up than one click up, because then your head in the clouds, and just so from, like, a design practice standpoint, you just want to, like, balance your boundary conditions between the, kind of, the logical rules and implementability conditions, inward, and the, like, contextual, sort of, integration and value creation one click up.

138

00:31:22.400 --> 00:31:22.990

Krzysztof Paruch: Right.

139

00:31:23.290 --> 00:31:25.940

Krzysztof Paruch: No, legit.

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00:31:26.450 --> 00:31:32.700

Krzysztof Paruch: Any other comments, maybe? Because the initial question actually was, whom are you building this market for, right?

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00:31:36.640 --> 00:31:43.880

Krzysztof Paruch: No, no, sorry, the actual question was, from which perspective are you looking at it? From the market design perspective, or from the participant's perspective?

142

00:31:50.950 --> 00:32:08.949

peterhacker: I just had a... one thought, about... it's probably more along the lines of, who are you building this market for, but I... I had an interesting conversation last weekend, about a prediction market that I saw... I forget where it got shared.

143

00:32:08.950 --> 00:32:17.959

peterhacker: Maybe it was in Block Science Slack? Probably not, though. But, it was a prediction market for DB train, arrivals.

144

00:32:18.000 --> 00:32:19.639

peterhacker: In Germany.

145

00:32:19.740 --> 00:32:34.879

peterhacker: And, like, whether they were on time or not, basically. And I was just talking to friends of mine, because we were... happened to be riding a train, and we were talking about how we might change our... or, like, could use the data as

146

00:32:34.880 --> 00:32:43.570

peterhacker: Like, to inform our, like, purchasing decision for which trains to buy for our trip or whatever. So that's kind of, I guess, an example of, like.

147

00:32:43.570 --> 00:32:52.390

peterhacker: You know, we wouldn't be participating in the market, trying to profit, but we would be producing or consuming based on that information, sort of, for your item, but not...

148

00:32:52.390 --> 00:32:56.790

Zargham: I was gonna say, that's a perfect example of the, you're not a free rider, you're buying a.

149

00:32:56.790 --> 00:32:57.280

peterhacker: Yeah.

150

00:32:57.280 --> 00:32:58.070

Zargham: Rain ticket.

151

00:32:58.070 --> 00:32:59.020

peterhacker: Exactly.

152

00:32:59.020 --> 00:33:23.989

Zargham: the broader ecosystem of, like, managing actual trains, there's no reason to consider someone who is using the prediction market outputs to inform, you know, train ticket purchasing, because they're not independent of the outward system. You're quite literally adjacent in that system. And so, again, not to be too pedantic about it, but, like, the immediate example is why I wouldn't consider an information consumer a freerider.

153

00:33:23.990 --> 00:33:26.819

Zargham: If they're otherwise connected into the system.

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00:33:28.320 --> 00:33:31.139

Krzysztof Paruch: Right. No, very interesting, very cool.

155

00:33:31.280 --> 00:33:40.430

Krzysztof Paruch: But let's maybe move on, we are, to make sure that we get enough time to cover all the information we want. Here, the next session, section is just,

156

00:33:41.300 --> 00:34:05.259

Krzysztof Paruch: just a revision of the mechanisms for anyone who wants to look at the document maybe later, and is not familiar with, kind of, like, how those mechanisms are specified. So, what we have are continuous double auctions and market scoring rules. Continuous double auctions are, just, a mechanism where you would have, participants submit orders,

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00:34:05.260 --> 00:34:19.639

Krzysztof Paruch: or market orders, or limit orders, so they can immediately execute, or they can place those orders that sit there for a while in an order book. And trades occur whenever there is a coincidence of wants.

158

00:34:19.659 --> 00:34:43.650

Krzysztof Paruch: And then you just basically calculate the price by looking at the midpoint of the best bid and best ask price. So you can say, for example, the limit order book stands at 95 and 90, so, like, obviously the market price could be 92.5, or you can look at it and say, what is the last executed trade? So this is also the market price. But this is basically how you set up the order book and the continuous double auction stuff.

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00:34:43.699 --> 00:34:55.930

Krzysztof Paruch: Which is something I will be probably exploring next. I haven't really looked into this much, but it's an interesting branch of research to really understand how the microstructure of an order book looks like.

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00:34:55.980 --> 00:35:19.370

Krzysztof Paruch: And then you have the market scoring rules, which are basically automated market makers, and they provide always liquidity, and you just choose one of the scoring rules, and then derive a cost function out of it, and then you say, okay, this is how much it costs you to move the price by selecting a different quantity of the inventory.

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00:35:19.370 --> 00:35:33.470

Krzysztof Paruch: Most prominent is the logarithmic market scoring rule, which has this definition, and then the prices are the gradient of this cost function, which gets you to each price for being

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00:35:33.470 --> 00:35:35.090

Krzysztof Paruch: They're defined exactly like this.

163

00:35:35.180 --> 00:35:58.659

Krzysztof Paruch: And, the interesting part is obviously that in the prices sum up to one and behave like probabilities, and this is exactly how much it costs one trader to... to move, the inventory from Q to Q_1 , okay? So this is just, a very high-level, summary for everyone who wants to, like, take the second D and then just, understand the full, situation. So what are we actually comparing?

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00:35:58.930 --> 00:36:14.700

Krzysztof Paruch: One, one, maybe one result is that you have some, some, losses, so you have a guaranteed, no, you have the maximum, maximum market loss that, that the initiator of the market, or the market maker itself.

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00:36:14.700 --> 00:36:26.350

Krzysztof Paruch: can incur, which is, defined by the liquidity parameter B , and, the natural logarithm of the number of outcomes. Like, if you... if you bet that,

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00:36:26.510 --> 00:36:34.560

Krzysztof Paruch: At 2 outcome market, yes or no, you would have the natural logarithm of 2, and then multiply it with the initialized liquidity parameter.

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00:36:34.880 --> 00:36:48.109

Krzysztof Paruch: Okay, so this is the setup, this is what we want to compare, and we want to compare performance within this market. But, coming back to your question initially, Jamshid, before we had the discussion,

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00:36:48.450 --> 00:37:04.890

Krzysztof Paruch: My interpretation of the economic analysis that I found is that it's mostly about, kind of like, designing the info... this... making this all to improve the quality of the market performance for the market, designer.

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00:37:05.910 --> 00:37:12.529

Krzysztof Paruch: And for all of the, kind of, like, I would say, even...

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00:37:12.580 --> 00:37:27.499

Krzysztof Paruch: not associated properties, for traders. So, so kind of, like, information aggregation itself. So, like, kind of, like, the quality of information aggregation, this is what, what, what, what, what, what is happening here. So, like, this is what, what those,

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00:37:27.600 --> 00:37:34.870

Krzysztof Paruch: experiments are trying to tackle. How to improve, or how to analyze the,

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00:37:35.000 --> 00:37:40.649

Krzysztof Paruch: Performance of a market with respect to how well and how quickly it can incorporate new information.

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00:37:41.400 --> 00:37:46.449

Krzysztof Paruch: And, such that, kind of, like, every observer can benefit,

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00:37:46.450 --> 00:38:09.750

Krzysztof Paruch: from it, because they see that this market has better prices, right? There is a lot of research also on, kind of, like, trader performance, but we are not talking about this. And we are not so much concerned about this yet. Like, we are now still in the shoes of the market designers, and we're trying to design a market, we're trying to implement it, we're trying to make a decision between when which microstructure makes more sense.

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00:38:09.860 --> 00:38:22.750

Krzysztof Paruch: And once we have this, we might be able to confront it against, kind of like, okay, what is the perspective of the user, and how will they behave, and how will they exploit the intricacies of the selected model?

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00:38:22.750 --> 00:38:23.899

Zargham: I have a thought on that.

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00:38:23.900 --> 00:38:28.179

Jamsheed Shorish: Well, one of the... oh, sorry, yeah, just very quickly to say that, just to...

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00:38:28.310 --> 00:38:31.140

Jamsheed Shorish: One of the ways in which you can actually have both

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00:38:31.380 --> 00:38:50.960

Jamsheed Shorish: perspectives is to not put an artificial cut between those two from the perspective of information aggregation, because definitions of market efficiency, i.e. that the market itself is reflecting the information and aggregating it in the best manner possible, or in the Pareto manner possible, benefits the users of the system.

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00:38:51.160 --> 00:39:09.360

Jamsheed Shorish: as well as the designer, whose goal, if you set it a priori as, I am here to design the most efficient market, is then fulfilled according to the same criteria. So there is really... because this is where all of auction theory came from, was the idea of how do we actually get efficient revelation of information

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00:39:09.360 --> 00:39:17.410

Jamsheed Shorish: And I think in this case, then, you can actually tie those two together as saying both users and the designers themselves, if that's a goal, would find that fulfilled.

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00:39:19.640 --> 00:39:20.580

Krzysztof Paruch: Right.

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00:39:20.580 --> 00:39:23.079

Zargham: So let me... Oh, go ahead, Chris, I'll let you.

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00:39:23.080 --> 00:39:25.040

Krzysztof Paruch: No, no, no, yeah, just... yes.

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00:39:25.590 --> 00:39:49.770

Zargham: I was gonna riff on this... the thing I was gonna say, I think, is related. The natural asymmetry between those two groups is often the, like, the depth of understanding of the market mechanisms themselves, and so, like, you know, when we think about optimal behavior on the part of a user, you sort of presume, often incorrectly, that there's a sufficient understanding of, and that it's incorporating an understanding of the market's mechanisms, the way that it's designed.

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00:39:49.770 --> 00:40:14.120

Zargham: the people designing it presumably have a high degree of expertise, the people consuming it generally have a more reinforcement learning-like pattern of behavior, which endogenizes less understanding of the market mechanisms themselves. And so one of the things I was interested in surfacing was the question of how the communication, demonstration, or the availability of a representation of the market mechanism might infect behavior. So.

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00:40:14.120 --> 00:40:35.470

Zargham: going to a lot of our model-based systems engineering work, the question becomes, if I have essentially the equivalent of a gym, right, in the AIRL sense, available that makes the model of the market mechanism available, both at the communications level, hey, how do we explain to users what the market mechanism is, and how easily do they

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00:40:35.470 --> 00:41:00.379

Zargham: interpret and integrate that into their thinking, and thus their behavior, whether it's a human, that's more like communications artifacts. As a machine, or an RL agent, that's more like building models that are actually... that can be easily incorporated into that agent, so that it has a world model that is essentially pre-built, assuming you want market... you want efficiency, and assuming that as the designer, you're expecting people to behave rationally.

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00:41:00.380 --> 00:41:07.499

Zargham: then I'm just thinking about, like, lowering the cognitive cost of behaving rationally, being a question of how...

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00:41:07.560 --> 00:41:31.819

Zargham: Effectively, you can communicate or serialize the mechanism itself, and within our work, which is very computational, what comes to mind for me is, like, can you take the mechanism, reduce it to a piece of software that is the mechanism in such a way that it is both easier to communicate, demonstrate, or allow a human to play with, and also, potentially, in the extreme case, be an easy-to-import submodel that

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00:41:31.820 --> 00:41:43.609

Zargham: that could be used both for RL training or direct incorporation into an RL agent to improve its rationality and practice, ergo meeting the assumptions of the market designer more readily.

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00:41:47.770 --> 00:41:49.660

Krzysztof Paruch: Okay.

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00:41:49.660 --> 00:42:14.579

Zargham: I don't know, Jamshit, if you have thoughts. I know we've been doing a lot of work on automation and computational work in other streams, so that might be a bit of a lift to jump for you, Chris, but I have this feeling that the problem... one of the problems here is that the market designer does tend to assume that the, you know, a certain amount of rationality on the part of the participants in ways that... there's two ways to deal with that. One is to relax your assumptions about how

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00:42:14.580 --> 00:42:28.759

Zargham: rational people are actually going to be, and using different requirements when you're designing with those weakened assumptions, or maybe to improve the ease of which a player behaves rationally, or

both.

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00:42:30.660 --> 00:42:50.370

Jamsheed Shorish: Yeah, I think just in terms of that, the latter case in which the information revelation is then something that can be provable or demonstrable is something which adds to that notion of, hey, we're going to make it easier on you to make informed choices, where you still have the agency to define what you mean by informed, and actually by what the mapping is between what information you receive and what choice you make.

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00:42:50.370 --> 00:42:50.860

Krzysztof Paruch: But.

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00:42:50.860 --> 00:42:57.699

Jamsheed Shorish: I think in that case, you kind of get that ability to say, this is the structure by which you can implement a version of, let's call it.

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00:42:58.120 --> 00:43:11.019

Jamsheed Shorish: What used to be called in the early 90s the literature on zero intelligence, which is that you just don't want to be made worse off, and so you're going to accept things that you can rationally assess as being an expected value term for yourself, making you better.

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00:43:11.020 --> 00:43:18.260

Jamsheed Shorish: And that's where the information clearing, the efficiency gains are the things that I think work in that regard and work to the advantage of the designer.

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00:43:18.260 --> 00:43:33.859

Jamsheed Shorish: And I think that's where you really want to be able to set up that triage of how do you decide the priorities of the market design problem in order to assume or to infer that rationality isn't going to be hitting some kind of a cognitive ceiling or cognitive block for individual assessment.

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00:43:34.020 --> 00:43:58.520

Zargham: That makes sense. So, the other thing I wanted to tie this back to is our, like... and again, this is a kind of ontological commitment of block science to separate mechanism from policy, and for the purpose of this discussion, this... we're kind of circling around,

can I give you the mechanism without the policy to improve your ability to deliver information to the system? So if I say to you, hey, I'm not going to tell you what you should care about, or why, or

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00:43:58.520 --> 00:44:21.770

Zargham: pretend like I know exactly what you should decide, that's the realm of your policy, right? The participant, the market user, is someone who has their own private preferences, objectives, beliefs, whatever, from which they derive their control actions, their market, their choice of action from the admissible action set, but the more that I give them a, like, well-tested, stable, germane-to-the-protocol.

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00:44:21.770 --> 00:44:46.409

Zargham: mechanism encoding, like, again, from a reinforcement learning standpoint, I actually dramatically lower the effort for you to perform optimally, given your information set, which is different from mine, given your preferences and beliefs, different from mine. It's the separation of the mechanism from the policy would allow me to argue that I can theoretically improve the, you know.

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00:44:46.410 --> 00:45:10.699

Zargham: let's call it the average rationality of the participants by equipping them with better, kind of, deterministic, if you did this in this circumstance, that would happen, while leaving their admissible action space open, so that they're the ones who are deciding their policies. We have, again, very GDS, like, we don't know what G of X is, we just want to make sure that they know what F of X comma U is, and what

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00:45:10.700 --> 00:45:34.340

Zargham: what the set U of U sub X is, and the more that we give them that machinery in deterministic, verifiable form, this is the state of the world, this is what will happen if you do that, then the behavior of a , again, either a computational or human actor should be materially improved, because you've lowered the computational cost of being rational.

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00:45:36.310 --> 00:45:39.769

Krzysztof Paruch: Right. I mean, to some extent,

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00:45:40.440 --> 00:45:54.859

Krzysztof Paruch: this is partially covered in one of the next papers that I want to present. I mean, like, at least the word zero

intelligence trader shows up. But, yeah, I see your point, so...

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00:45:55.440 --> 00:46:08.290

Krzysztof Paruch: Let me maybe move on, and... and just kind of, like, for the next, like, 5 and 5 minutes, talk through briefly the two papers that I have for today, and then we have, like, 5 to 7 minutes, remaining time for discussions.

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00:46:08.290 --> 00:46:17.860

Krzysztof Paruch: Okay, so this is the simulation model, Klingon and Meyer. What they do is they use this experiment-based micro-validated multi-agent simulation based on Hansen's initial lab experiment.

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00:46:18.000 --> 00:46:23.789

Krzysztof Paruch: And, yeah, what they want to do is to isolate mechanism-related effects on prediction market performance.

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00:46:24.000 --> 00:46:46.560

Krzysztof Paruch: Okay, so what they do, three different things, lab experiment data used to calibrate and validate the model from Hansen, then they set up this agent-based simulation, and then they do experimental design analysis using a factorial simulation setup, so basically going through all of the combination possibilities and running different runs to see how this performs, and then making statistical analysis of the results.

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00:46:46.750 --> 00:46:49.679

Krzysztof Paruch: So, they test 5 hypotheses.

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00:46:49.970 --> 00:47:03.829

Krzysztof Paruch: But they are focusing on number of trades, accuracy level, and standard deviation of price, so those are the three topics that they focus on, and they say, the market scoring rule has a higher number of trades than the continuous double auction.

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00:47:03.920 --> 00:47:11.069

Krzysztof Paruch: The market scoring rule is more accurate than the continuous double auction, but they define two different sub-hypotheses here.

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00:47:11.190 --> 00:47:14.229

Krzysztof Paruch: Especially in the presence of random strategies.

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00:47:14.580 --> 00:47:17.489

Krzysztof Paruch: Which is kind of like the thing that I alluded to in the beginning.

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00:47:17.660 --> 00:47:32.279

Krzysztof Paruch: And, however, in the case of extreme information distribution, so, for example, if there are people who know stuff that is, like, it's on the boundaries, then the continuous double auction or the book model performs better.

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00:47:32.360 --> 00:47:39.500

Krzysztof Paruch: And what they also say with respect to accuracy is that the market scoring rule has a lower standard deviation of the final price than the CDA.

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00:47:40.030 --> 00:47:44.950

Krzysztof Paruch: Okay, so this is how this looks. We won't have time to discuss it in detail, but this is

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00:47:44.980 --> 00:47:48.629

Krzysztof Paruch: Yeah, just, like, different formulation of what is just said above.

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00:47:48.630 --> 00:48:06.780

Krzysztof Paruch: The simulation model is, based on the laboratory experiment that they've had. So, they have 12 traders with 200 monetary units, and there are 2 stocks that can be traded, and they... those two stocks have three different payoffs. So, at the outcome, what can happen is, like, 0, 40, or 100 is paid out for each stock.

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00:48:06.780 --> 00:48:08.640

Krzysztof Paruch: And,

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00:48:08.780 --> 00:48:25.240

Krzysztof Paruch: the interesting part is that each trader knows one value that cannot be true. So, like, they tell everyone one of those three cannot just be the realized outcome. No one knows all of them, knows the full information, but the market knows the full information. So, like, the correct value can be inferred collectively.

224

00:48:25.450 --> 00:48:41.869

Krzysztof Paruch: And then there are different agent strategies, so those are the zero intelligence EV traders, then you have the normalized traders with expected value and last price, they have fully random traders and adaptive learning traders, and they,

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00:48:42.040 --> 00:48:45.589

Krzysztof Paruch: Set up those strategies, in different models.

226

00:48:45.590 --> 00:49:09.660

Krzysztof Paruch: Then they take those different parameters of the experimental setup and run through all combinations. So how this looks like is exactly like this. So you look at the mechanism, the strategy, information distribution, steps, number of agents, interest stocks, and agent money, and what you see here in bold are the numbers that Hanson selected. So, like, they are trying to kind of, like, sweep around those numbers and see how the mechanisms would have performed.

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00:49:09.660 --> 00:49:15.490

Krzysztof Paruch: Like, yeah, maybe kind of, like, extend the laboratory experiment through numerical simulations.

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00:49:15.670 --> 00:49:33.420

Krzysztof Paruch: How this looks, as an example run, is you see the CDA, the order book jumps more heavily, because, like, you have kind of, like, information coming in, and then you have bigger, bigger jumps and bigger price adjustments. The market scoring rule, the AMM, is continuous, so, like, you have a smoother trading going on.

229

00:49:33.710 --> 00:49:45.960

Krzysztof Paruch: And you also see this when you evaluate box plots of multiple simulation runs. So, like, the scoring rule, again, is centered around the expected value, and the CDA has higher variance.

230

00:49:46.500 --> 00:50:04.480

Krzysztof Paruch: What are the results? The results are that Hypothesis 1 is supported or accepted. You see that the logarithmic market scoring rule has, in the default setting, more than 3x more trades, but in the... in, like, overall settings, around double.

231

00:50:04.540 --> 00:50:14.620

Krzysztof Paruch: twice as many. What they say is because, like, it's, you can... you have continuous liquidity, so, so you can always trade,

and this is why you have more trades.

232

00:50:14.820 --> 00:50:31.119

Krzysztof Paruch: On the accuracy level, they are not, certain, so, like, they reject the hypothesis, because there is no consistency between the default setting and all settings. So, like, you see here that you would have, yeah, more trades here.

233

00:50:31.160 --> 00:50:37.280

Krzysztof Paruch: But... yeah, no, sorry, you would have, what is it, just, like, water testing? A standard...

234

00:50:37.440 --> 00:50:40.649

Krzysztof Paruch: deviation, I guess, right? So, like, you have...

235

00:50:41.400 --> 00:50:43.830

Krzysztof Paruch: No, trades, number of trades, I guess.

236

00:50:44.090 --> 00:51:05.850

Krzysztof Paruch: Sorry, I'm mixing this up now, but definitely Hypothesis 2 is rejected, but Hypothesis 2A is supported, and 2B is supported. So what this means is that if you have random strategies, then LMS error is more accurate, and if you have extreme information distribution, then the order book is more accurate.

237

00:51:05.970 --> 00:51:14.219

Krzysztof Paruch: This is what you see here. And for the price volatility, what they say is,

238

00:51:14.410 --> 00:51:21.870

Krzysztof Paruch: It's the same, so, support for hypothesis 3, market scoring rule has lower standard deviation than the final price.

239

00:51:22.300 --> 00:51:26.579

Krzysztof Paruch: And this is a very extreme, so to, so to say.

240

00:51:26.970 --> 00:51:49.410

Krzysztof Paruch: And this is how they basically wrap up, right? So, like, they do those tests based on simulation experiments and say, okay, we had some hypotheses, and we support or reject them, and they provide some explanations for this. For example, in the accuracy case,

they say that the market scoring rule prices depend on all trades.

241

00:51:49.410 --> 00:52:14.169

Krzysztof Paruch: So this is a recurring argument, where they say, like, okay, this incorporates kind of, like, the whole history of the trade, compared to the order book, where you can simply do one trade, and with one trade, you can move the price from one extreme to the other, which is also why, in the extreme distribution case, it's very easy to use the order book and say, hey, okay, we just completely make, like, one jump to the other extreme.

242

00:52:14.170 --> 00:52:19.309

Krzysztof Paruch: Because now some information was revealed, and we need much less trades to achieve this result.

243

00:52:19.660 --> 00:52:20.700

Krzysztof Paruch: Okay?

244

00:52:20.830 --> 00:52:33.950

Krzysztof Paruch: So let's go over to the next paper, because we don't have much time. This is Karimi and Dimitrov, 2024. They say... they ask subsidize or not to subsidize. I mean, this is coming from the fact that you have to subsidize

245

00:52:33.950 --> 00:52:58.909

Krzysztof Paruch: the automated market makers, right? So you have to... you have to provide an initial endowment, you have to provide liquidity, and this is costly, and... or some other form of subsidy to, incentivize traders to reveal information. You have to do it, right? And this is also the case why the market scoring rule AMMs, they have this bounded loss, where they say, like, there is a maximum amount

246

00:52:58.910 --> 00:53:23.750

Krzysztof Paruch: amount of money that can be lost from trading, because adverse selection, right? Like, the AMM always trades, regardless what the information is, regardless of what traders know, they accept all the trades, and therefore there is some liquidity at stake, even. But this is the price of, like, running this market. This is the price of revealing information. So, the question is, do you want to subsidize the market? Is it worthy of

247

00:53:23.750 --> 00:53:38.870

Krzysztof Paruch: kind of like the subsidy to reveal information, to

even create a market in the first place? Or can you, you know, get away without such a subsidy and just deploy CDA markets and say, okay, we will get the same information for free?

248

00:53:39.150 --> 00:53:40.280

Krzysztof Paruch: Okay?

249

00:53:40.330 --> 00:53:47.090

Krzysztof Paruch: Okay, on a high level, the result is, and I'm gonna make it very quick, is that there is a threshold.

250

00:53:47.090 --> 00:54:03.549

Krzysztof Paruch: So, like, until a particular threshold, you can, logarithmic markets or scoring rule markets perform better, and above this threshold, information is already, you know, dispersed enough, and the activity is high enough, so, like, you can get the information for free.

251

00:54:03.590 --> 00:54:08.680

Krzysztof Paruch: Where do I have this? I have this,

252

00:54:08.940 --> 00:54:33.599

Krzysztof Paruch: somewhere here. So they state two hypotheses. They say, the market scoring rule should be better than the continuous double auction in low trade markets, but in high trade markets, what they assume first is that they should be equal, but they reject this hypothesis, and then they clearly say, no, it's actually the other way around, so the CDA is better. And how do they arrive at those results? What they do is.

253

00:54:33.600 --> 00:54:34.240

Krzysztof Paruch: Hmm...

254

00:54:34.240 --> 00:54:51.429

Krzysztof Paruch: They say, we have also two run... two markets running simultaneously, and those are real markets that have been actually deployed in practice, where people have been trading on one particular event on two different market microstructures.

255

00:54:51.430 --> 00:55:11.120

Krzysztof Paruch: So there is this in-trade and this IPredict applications. Back then, in 2008, this was completely, you know, off-

chain, but there were those applications where people could have traded, and they were run simultaneously, and they found, actually, identical markets, and they were able to do a statistical analysis.

256

00:55:11.120 --> 00:55:12.510

Krzysztof Paruch: of the performance.

257

00:55:12.590 --> 00:55:28.960

Krzysztof Paruch: between those two markets. So, first, pairing markets, then, testing if there is a shared efficient per price. So, if there is a co-integration, or integration of order one, of those time series, underlying this market.

258

00:55:29.270 --> 00:55:48.559

Krzysztof Paruch: And then you just normalize this and compare. So they do this via this VSEA model from Hasbrook, where they define those predicted price shares, those changes in predicted price shares, set them up accordingly, and you see here that the

259

00:55:48.560 --> 00:56:00.710

Krzysztof Paruch: First term represent the long-term relationship between the two market prices, the second and the third term, so those two represent the short-term relationship, and the third term, which is the epsilon, represents some error.

260

00:56:00.710 --> 00:56:15.540

Krzysztof Paruch: And based on that, they make some calculations. So, like, they compute two price discovery measures, which is the component share, and then they compute the information share, and then they plot those shares on regression.

261

00:56:15.540 --> 00:56:33.709

Krzysztof Paruch: And see, when does it make sense to say, like, when is the threshold under which the component share or the information share of those markets are relatively higher or lower, and this is exactly the case where they recommend to switch.

262

00:56:33.710 --> 00:56:37.679

Krzysztof Paruch: Between the market microstructures. I hope this was a good...

263

00:56:37.830 --> 00:56:50.640

Krzysztof Paruch: summary in sake of time. So... so what they say is, in thin markets, the scoring rule has advantage. We knew this already. In thick markets, CDA has the advantage.

264

00:56:50.640 --> 00:57:10.379

Krzysztof Paruch: we didn't know this, because the, like, literature consensus was that it's kind of actually equal, and then there is a crossover that depends on trade volume, and you can derive it. So, one recommendation, one practical recommendation that they give is, before installing a market, you should kind of, like, try to estimate the number of trades, and then you can decide on the microstructure.

265

00:57:10.770 --> 00:57:29.049

Krzysztof Paruch: And then finally, we have Artsunov et al. Yeah, I shared it yesterday, but I didn't really get to it. What they do there is they run this whole long experiment where they... where they let people participate in both markets, and then they calculate the Bria score exposed, so they try to see

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00:57:29.210 --> 00:57:36.520

Krzysztof Paruch: If this... how well their predictions were compared to the realized outcome. Okay, jump sheet, 6 minutes.

267

00:57:39.150 --> 00:57:52.699

Jamsheed Shorish: Sure, I mean, I want to make sure that you feel like you've given yourself enough time. My comments are really optional, they really were more along the lines of wrapping up kind of an arc. If you want to continue to dig deeper in the rest of what you've already got, please feel free to do so, otherwise I can quickly jump in with a couple comments.

268

00:57:53.110 --> 00:58:09.560

Krzysztof Paruch: There's not much really to say. I mean, like, I can... like, this section I already presented above, because it is just a literature, list, and then, like, the only thing that I could really talk about are those equations, but I don't think that this, like, benefits anyone if we do this now, like, for the next 5 minutes.

269

00:58:09.770 --> 00:58:15.440

Krzysztof Paruch: I believe it makes more sense to just discuss or anchor it into something that is more practical.

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00:58:17.610 --> 00:58:22.390

Jamsheed Shorish: Okay, very good. Let me, I'll just quickly take some of the remain... take the remaining time just to give...

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00:58:22.390 --> 00:58:38.889

Jamsheed Shorish: a very quick overview of some comments on a relationship with... between this and a way in which we can view this. It's sort of, like, leading to the comments that we had in the discussion that we already had about, like, what's it for, how is it useful, to kind of couch it formally more within the model-based systems engineering perspective.

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00:58:39.080 --> 00:58:49.880

Jamsheed Shorish: If anything like that, so basically the, if everybody can see that slide, it's just really just a comment on how we can kind of view this prediction market evaluation process with regard to the lens of MVSE.

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00:58:49.970 --> 00:58:53.299

Jamsheed Shorish: And the idea here really is to associate,

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00:58:53.300 --> 00:59:10.330

Jamsheed Shorish: different subsystems, different associated mechanisms with types of evaluation criteria that are consistent and interpretable. That's what we really want to make sure we understand that these things are consistently measured. You can see that there were a spread of different things that were actually used as measurements of whether or not the system is actually good in quotations.

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00:59:10.330 --> 00:59:12.410

Jamsheed Shorish: And what we want to look at is this idea

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00:59:12.410 --> 00:59:25.970

Jamsheed Shorish: always from a systems engineering point of view, first of understanding how the interface composability works. How can you measure individuals connected to each other? So, for example, the systems that come out of the approach that... of the papers you've, you know, put together.

277

00:59:25.970 --> 00:59:41.149

Jamsheed Shorish: Market mechanism subsystem, participant management subsystem, transfer subsystem, scoring subsystems, how do they all fit together so that you can measure things consistently? In addition, of

course, you've got the aggregation question. Once you've got all of these different subsystems, how do you get system performance?

278

00:59:41.150 --> 00:59:58.699

Jamsheed Shorish: Like, when you take a look at the market mechanism subsystem, how it aggregates with participation, up to the scoring subsystem, which ultimately provides a measure. And that key here really is just to develop or adopt trade-off measures. You really want to figure out which ones are actually telling you something about composability, and understand which ones seem to be more or less effective.

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00:59:58.720 --> 01:00:03.779

Jamsheed Shorish: And just to couch that within the terminology of what we've looked at before in MBSC,

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01:00:03.780 --> 01:00:19.700

Jamsheed Shorish: What we've seen from the paper is that we've got, very loosely speaking, a series of performance measures. We've got accuracy, we've got the variance of behavior, we've got information composition, decomposition, how rapidly it's being assimilated, for example. Each of the papers kind of has a different bent or

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01:00:19.700 --> 01:00:25.650

Jamsheed Shorish: or let's say, emphasis on one over the other, although they're not mutually exclusive. These all support, though.

282

01:00:25.710 --> 01:00:35.530

Jamsheed Shorish: the end-state MBSC measures of effectiveness on those. These are the questions that are answered. Is the system doing what it's supposed to be doing from the user's perspective?

283

01:00:35.640 --> 01:00:49.580

Jamsheed Shorish: And then the quantifications themselves are actually... the performance measures are MOPs, measures of performance, which says how well is the system doing what it's supposed to do from the developer's perspective, or the implementer's perspective, who's actually made those and performed those quantifications.

284

01:00:49.580 --> 01:01:04.409

Jamsheed Shorish: And those two, in this particular case, they focus upon the things that are the conceptual and the functional attributes of the implementation. What are we thinking about? Why is this here?

And how is it supposed to do things without being specific about what it is that's actually being implemented?

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01:01:04.410 --> 01:01:06.389

Jamsheed Shorish: And of course, to lead on to that.

286

01:01:06.390 --> 01:01:16.900

Jamsheed Shorish: you've got to also understand, well, okay, is this being constructed correctly? Are we putting things together correctly? These are all the building block pieces that each of the pavers simply assumes as the model question.

287

01:01:16.900 --> 01:01:28.540

Jamsheed Shorish: But for us, it does become important, even though it wasn't emphasized before, about how parameters or models are selected when implemented in the actual market. Parameter boundary conditions, confidence intervals, aggregation functions.

288

01:01:28.540 --> 01:01:31.660

Jamsheed Shorish: Agent strategies, what kinds of agents, and how do they interact?

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01:01:31.660 --> 01:01:56.449

Jamsheed Shorish: all of these things are in MBSC, put into a perspective around technical performance measures, or TPMs, asking how well is a subsystem or a subsystem element fulfilling its technical requirements? And of course, that focuses a lot upon the logic that you build. You have a black box for an agent, you open that black box up, you put some behavior in there, that's now a logical subsystem, logical attributes of that agent, the physical implementation, how do you actually make this thing work?

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01:01:56.950 --> 01:02:03.740

Jamsheed Shorish: Those are the pieces that then focus upon that when they're building the system from an engineering point of view. And so, just from the point of view of

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01:02:04.000 --> 01:02:22.690

Jamsheed Shorish: couching that within the context of MBSC, these are some ways in which you can link some of the academic work that was just presented into a structure to actually build if you were to actually task this as an engineering project. And I put at the end there just a quick one that I like very much as an engineering

handbook. The NCOSI one I can't get.

292

01:02:22.690 --> 01:02:34.879

Jamsheed Shorish: because it's behind a paywall, but the NASA Systems and Hearing Handbook is quite... is open, it's freely available, and it has a very nice description of this to be able to kind of, like, get a perspective on building this as if it were a project, so...

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01:02:34.880 --> 01:02:38.029

Jamsheed Shorish: That's the comment. It was just to be able to kind of, like.

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01:02:38.030 --> 01:02:44.050

Jamsheed Shorish: tie something together within regards to the content of what we've been working on also at Block Science.

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01:02:44.580 --> 01:02:45.450

Jamsheed Shorish: Thanks.

296

01:02:46.380 --> 01:02:59.260

Krzysztof Paruch: Cool. Perfect, thank you. No, this is amazing. I mean, this is obviously tying together all of the different aspects that are needed in order to... not only to build, but also to, monitor the system once it's live, so...

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01:02:59.890 --> 01:03:04.709

Krzysztof Paruch: Very, very cool. So it's basically the adjacent, data

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01:03:04.920 --> 01:03:08.900

Krzysztof Paruch: Collection and analysis pipeline that needs to be developed.

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01:03:10.770 --> 01:03:13.480

Krzysztof Paruch: Next to the actual market implementation.

300

01:03:16.300 --> 01:03:16.970

Jamsheed Shorish: Nice.

301

01:03:17.150 --> 01:03:21.059

Jamsheed Shorish: Well, great, I think that... I guess we are at time. I think,

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01:03:21.400 --> 01:03:31.590

Jamsheed Shorish: Yeah, I think, Zargum, you had a comment in the channel about having, maybe, connecting with Chris with regard to looking at tooling, moving beyond, perhaps, HackMD, definitely makes sense.

303

01:03:31.590 --> 01:03:44.110

Zargham: For the moment, since we're at time, Chris, grab that link and poke around it, because, like, honestly, it's a public repo, so you can read it, play with it, pull it, but I only recently found learning mostly from

304

01:03:44.110 --> 01:04:09.060

Zargham: Rohan's work, how... how powerful and easy it can be to make notebooks, and particularly for reproducing analytic derivations and stuff using SymPy. And so, I just recommend that you poke around that and think about it. We can chat about it on another call, we could have a session dedicated to it if we want to in the future, but I just... given what you're doing, it... it just dawned on me that, like, some of what I've learned

305

01:04:09.060 --> 01:04:12.070

Zargham: Recently might be relevant to you vis-a-vis tooling.

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01:04:13.100 --> 01:04:20.780

Krzysztof Paruch: No, this is amazing. Final question before we close. Any, I mean, recommendations directionally, or recommendations, wishes?

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01:04:21.040 --> 01:04:31.400

Krzysztof Paruch: For the next session, I mean, like, I'm... maybe I will follow your advice, maybe I will do something else, but I was just wondering what the... what the interest of the group would be, given what we've talked so far.

308

01:04:38.500 --> 01:04:57.340

Zargham: I've been happy, I mean, it's been a really great, like, session. I don't know what criteria you're currently using to choose your scopes, but for the most part, they've been, like, productive, sessions, so we can take it offline and chat about it in the channel

if you need guidance in terms of what to do the next session on.

309

01:04:58.780 --> 01:05:00.520

Krzysztof Paruch: Nice. Cool. Perfect.

310

01:05:01.290 --> 01:05:02.939

Krzysztof Paruch: That's all from my side.

311

01:05:03.650 --> 01:05:15.779

Jamsheed Shorish: All right. Well, great. Thanks again, Chris, for the time. Really, as always, fascinating. It's great to see the progress, and I look forward to the next iteration, be it in this subject matter or in a new direction. Thanks, everyone.

312

01:05:15.780 --> 01:05:17.490

Krzysztof Paruch: Perfect, thank... thank you, bye-bye.

313

01:05:17.490 --> 01:05:18.690

David Sisson: Thanks, Chris, thanks all.

314

01:05:18.690 --> 01:05:19.740

Jamsheed Shorish: Thank you, bye.

315

01:05:19.740 --> 01:05:20.479

peterhacker: See ya.